Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Parallel axis

Meaning in my words How would you explain parallel axis to a friend?	**How can you tell?** If downhill is positive, what sign does the along-ramp part of gravity have?
Example or sketch Sketch axes for a ramp and mark the parallel axis.	**Non-example and why** A partner uses the horizontal floor as the parallel axis. Which surface should they follow?

Perpendicular axis

Meaning in my words How would you explain perpendicular axis to a friend?	**How can you tell?** If the positive axis points away from the ramp, what sign does the normal force have?
Example or sketch Draw the perpendicular axis for a block on a tilted ramp.	**Non-example and why** A student draws that axis straight up on a tilted ramp. Why would you rotate it?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Weight components

Meaning in my words What are the two components of weight on a ramp?	**How can you tell?** Which part of weight pulls downhill, and which presses into the ramp?
Example or sketch Draw weight and its two parts. Which arrows should enter the two ramp-axis equations?	**Non-example and why** A student adds weight and both of its components as three forces. What got counted twice?

Constraint

Meaning in my words How would you explain constraint to a friend?	**How can you tell?** A block stays on a straight ramp. What is its acceleration perpendicular to the surface?
Example or sketch How would you use that fact to find the normal force when gravity and the surface are the only forces?	**Non-example and why** Does zero perpendicular acceleration mean zero downhill acceleration?

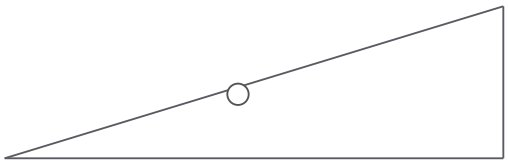
Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER

Name _____

Draw the real weight once, then draw its two components in a separate box. Circle the component along the ramp.

Real forces



Components

My marks mean

One pull, two components

16

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Represent one force without counting it twice.

CHOOSE ONE JOB TOGETHER

NAVIGATION ENGINEER ☐

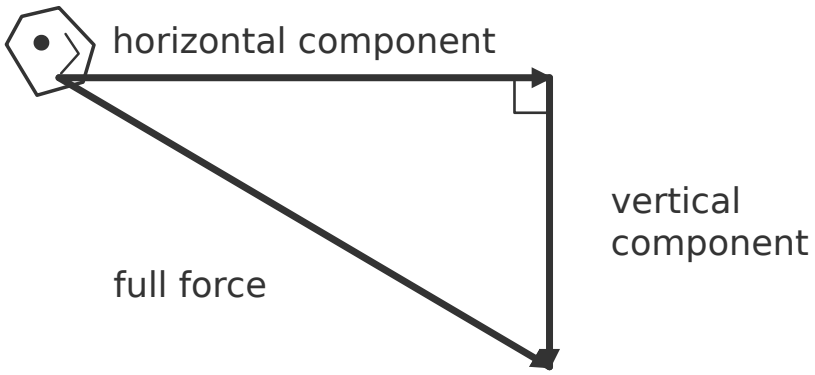
Prepare the component view for the calculator. Mark the full-force arrow that these components replace.

MAP GRAPHICS EDITOR ☐

Show the full force and its component view. Write a caption that prevents counting the same pull twice.

Point to a clue. Keep your first marks; add the update using the color key.

A navigation graphic lists a gravitational force AND its two components as three separate pulls on XJ.



01 OUR FIRST CALL

Which arrows should enter the force sum? Explain your choice.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Check the calculator request. What correction would you send before it runs?

Switch speakers. If XJ's velocity points horizontally, which force component changes its speed at that instant? Why?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks